Character accuracy conceals numerical error: measuring silent value corruption in a clinical document pipeline

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Abstract

Clinical decision support systems increasingly read patient data from document images, where optical character recognition sits between the page and every downstream inference. Reported accuracy for that stage is usually character-level, and a laboratory value is not a character. We measured the image-to-value path of a retrieval-based clinical decision support prototype under controlled image degradation, using synthetic reports carrying exact machine-readable ground truth for 13 analytes, with the document as the experimental unit across 10 independent reports and 7 scan resolutions. At 100 dpi, character accuracy remained at 88.9% while 32.3% of laboratory values were misread and a further 10.0% were not read at all. Of 188 corrupted values, 89 (47.3%) carried the same abnormality flag as the true value, leaving the error invisible to the alerting layer. We identify the mechanism: because abnormality flags are threshold quantisations, corruption of a value that was already outside its reference range is substantially more likely to preserve the flag than corruption of a normal value (61.7% against 22.1%, Fisher exact $p = 1.6 \times 10^{-7}$, odds ratio 5.68). Fourteen of fifteen corrupted troponin readings were hundredfold overstatements that retained the *Critical* flag. A physiological plausibility bound rejected 104 of 188 corruptions and recovered 33 of the 89 silent ones, failing entirely for analytes whose plausible range is wide relative to their reference range. Substituting a large language model for the deterministic extractor repaired nothing. These findings come from synthetic documents and one OCR engine, and they bound no claim about real clinical performance; what they do establish is that character-level accuracy is not a safety metric for pipelines that reduce measurements to flags.

1 Introduction

A clinical decision support system that reads a patient record from a scanned page performs a chain of inferences before any model reasons about the patient. Pixels become characters, characters become a labelled value, the value is compared against a reference range, and the resulting flag reaches the clinician or the generator. Each step can fail, and the failures do not announce themselves in the same way.

Work on retrieval-augmented generation in healthcare has concentrated on what happens after a value exists: whether the right passage is retrieved, whether the answer is grounded, whether a citation points where it claims to [1, 3, 4]. That literature assumes the number arriving from the document is the number printed on it. When the document is a photograph or a fax, the assumption is doing work that nobody has measured.

Optical character recognition is normally reported through character error rate, and character error rate treats every symbol as equivalent. A decimal point is one character. In `5.8 mEq/L` it is also the difference between a potassium concentration requiring monitoring and one incompatible with life. A pipeline can therefore post a high character accuracy and still deliver a wrong number, and the number is what the clinical logic consumes.

We report two experiments on this path. The first sweeps 32 degradation conditions on a single document to establish which corruptions occur. The second takes the document as the experimental unit

across 10 independent reports and 7 resolutions, and measures how often values are misread, how often they are lost, and what the system's own flagging layer does with the result.

Our contributions are the following. We quantify the dissociation between character accuracy and value accuracy in a clinical extraction pipeline. We give a taxonomy of what corruption does to an abnormality flag, and show that the most common outcome is no change at all. We identify the mechanism behind that outcome, which is threshold quantisation, and show it is predicted by whether the true value was already abnormal. We evaluate a physiological plausibility bound and characterise where it systematically fails. We report that replacing the deterministic extractor with a language model repaired no error and introduced none.

2 Related work

Retrieval-augmented generation has been proposed widely for clinical question answering, and recent reviews survey its use across healthcare tasks [1, 3]. Systems such as ClinicalRAG retrieve over heterogeneous clinical knowledge [4], and question-answering corpora such as MedQuAD provide evaluation material [2]. These evaluations measure retrieval and generation quality on text that is already text.

Clinical natural language processing generally begins from already transcribed records, and OCR quality is reported, where reported at all, as a character or word error rate. We are not aware of a study that measures how character-level degradation propagates into extracted numerical values and then into the abnormality flags that clinical logic acts on, which is the gap this work addresses.

3 Methods

3.1 Documents and ground truth

Each synthetic report is generated with a machine-readable truth file recording the exact value, unit, printed reference range and expected flag for 13 analytes: fasting glucose, HbA1c, haemoglobin, ferritin, white cell count, platelets, sodium, potassium, blood urea nitrogen, creatinine, eGFR, troponin I and LDL cholesterol. The panel was chosen to span several orders of magnitude and several decimal conventions, so that a lost decimal point produces an error of different size depending on the analyte.

No real patient data were used at any stage.

3.2 Degradation conditions

Documents are rendered from PDF at a specified resolution and then degraded. The condition set covers resolution (300, 200, 150, 120, 100, 85 and 72 dpi), Gaussian blur ($\sigma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$), JPEG compression (quality $\in \{90, 70, 50, 30, 15\}$), rotation ($\{0.25, 0.5, 1.0, 2.0, 5.0\}$ degrees), additive noise ($\{5, 10, 20, 30, 50\}$), and five composites intended to resemble real capture paths: a good scan, an office scan, a phone photograph, a poor fax and a poor photograph.

Stochastic conditions are run with five seeds. Experiment 1 therefore comprises 24 single-seed conditions and 8 conditions at five seeds each, giving 64 runs and, across 13 analytes, 832 observations on one document.

3.3 Extraction, flagging and the plausibility bound

Degraded images are passed to Tesseract (`-psm 6`). A deterministic parser locates each analyte by label and extracts the first numeric token on the line.

The flagging logic is transcribed, rather than imported, from the prototype's `clinical_parser` module so that the experiment runs without the application stack. The transcription preserves a property of the deployed system worth stating plainly: its reference ranges are fixed, are not age- or sex-aware, and are not the ranges printed on the document. The system flags haemoglobin outside 12.0–17.5 g/dL while the generated report prints 12.0–15.5 g/dL.

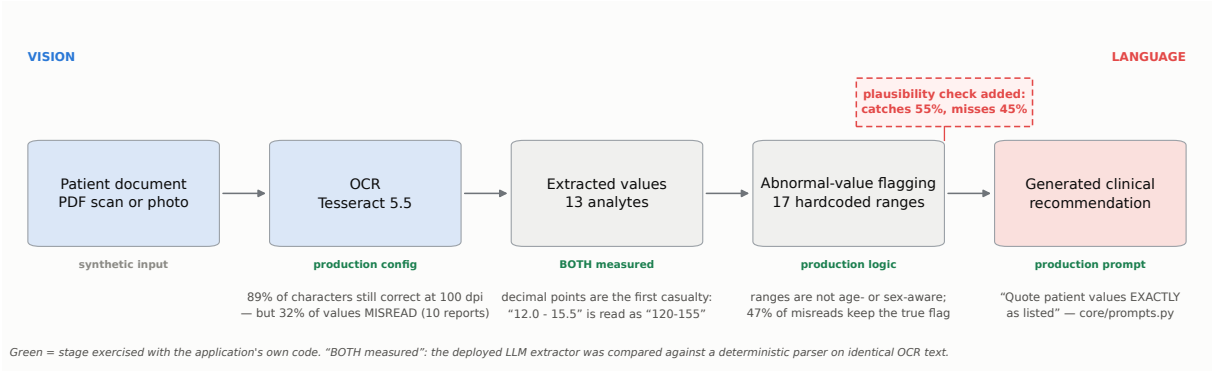


Figure 1: The measured path, with one observed case. A potassium concentration of 5.8 mEq/L printed on the page is extracted as 58.0 and carries the same *High* flag as the true value. Stages marked in green were exercised using the prototype's own code. The potassium case is one observed instance from the single-document sweep and is not a frequency. Whether a corrupted value changes a generated recommendation was not tested.

We additionally implement a physiological plausibility bound, a generous envelope of values possible in a living patient, and ask whether it would have rejected each corruption. This envelope is our own construction for this study and is not a clinical standard.

3.4 Experimental unit and the calibration gate

Thirteen analytes read from one page are not thirteen independent observations, since they share a rendering, a scan and an OCR pass. Experiment 2 therefore treats the document as the experimental unit: proportions are computed within each of 10 independent reports and summarised across them, and every dispersion we report is a standard deviation between documents rather than a confidence interval.

No result is reported for a document unless its clean 300 dpi render returns all 13 values correctly. This gate separates degradation effects from generation or parsing faults. All 10 documents passed.

Experiment 2 comprises 10 documents $\times$ 7 resolutions $\times$ 13 analytes, giving 910 observations.

3.5 Outcome definitions

Each reading is *correct*, *misread* (a value was emitted and it differs from truth) or *unread* (no value was emitted for that analyte). Misread and unread are different failures with different consequences and we never pool them.

Character error rate is Levenshtein distance between the whitespace-normalised reference text and the OCR output, divided by the length of the reference.

4 Results

4.1 Which degradations corrupt values, and which destroy them

Experiment 1 sweeps 32 conditions on one document to establish which failure modes exist before committing the replicated design to a single axis. Table 1 gives the outcome by degradation family.

The families separate by failure mode. Resolution is the dominant source of wrong values. Noise and the composite capture paths mostly prevent the analyte's label from being located, so the value is absent rather than wrong. JPEG compression and small rotations produced no error of either kind at the levels tested.

This distinction matters for risk. An analyte that is not found is an absence a caller can detect. A misread analyte is a confident wrong answer. Because resolution is the condition that most readily produces the second case, Experiment 2 replicates the resolution axis across 10 independent documents.

Table 1: Experiment 1: outcome by degradation family, 832 observations on one document. *Corrupted* means a wrong value was emitted; *label not found* means the analyte's line was not located at all. JPEG compression to quality 15 and rotation to five degrees produced neither outcome.

Family	Observations	Corrupted (%)	Label not found (%)
Baseline (300 dpi)	13	0.0	0.0
JPEG quality 90–15	65	0.0	0.0
Rotation 0.25–5°	65	0.0	0.0
Noise 5–50	325	1.5	20.0
Blur σ 0.5–3.0	65	9.2	0.0
Composite capture paths	221	6.3	56.1
Resolution 200–72 dpi	78	23.1	10.3

Table 2: Character accuracy and value outcomes across the resolution sweep. Each cell is the mean over 10 independent documents, with the standard deviation between documents in parentheses. Misread means a wrong value was emitted; unread means no value was emitted.

Resolution	Character accuracy (%)	Values misread (%)	Values unread (%)
300 dpi	98.10	0.00	0.00
200 dpi	98.02	1.54 (3.24)	0.00
150 dpi	96.67	4.62 (5.38)	0.00
120 dpi	96.12	12.31 (9.73)	0.00
100 dpi	88.88	32.31 (9.46)	10.00 (5.19)
85 dpi	80.83	49.23 (13.66)	14.62 (5.68)
72 dpi	56.53	44.62 (21.08)	41.54 (27.43)

4.2 Character accuracy and value accuracy separate

Table 2 gives both measures across the resolution sweep. At 300 dpi no value was misread in any of the 130 analyte readings. Degradation then affects the two measures at different rates. Between 300 dpi and 100 dpi, character accuracy falls by 9.2 percentage points while the misread rate rises from zero to 32.3%.

Below 85 dpi the misread rate stops rising while the unread rate more than doubles. Paired by document between 85 dpi and 72 dpi, the misread rate fell for four documents, rose for three and was unchanged for three (mean change -4.6 percentage points, SD 20.9). The data are consistent with values ceasing to be read rather than being read wrongly, and they do not establish a decrease.

4.3 What corruption does to the flag

Across Experiment 2, 188 of 910 readings were misread. Table 3 classifies each by what happened to the abnormality flag. The largest category is no change: 89 corruptions (47.3%) produced a value that carried the same flag as the truth. For those readings the error is invisible to any component that consumes the flag rather than the value.

4.4 Already-abnormal values hide their own corruption

The silent category is not distributed at random. Splitting the 188 corruptions by whether the true value was already outside its reference range gives 61.7% silent among the 120 corruptions of abnormal values, against 22.1% among the 68 corruptions of normal values (Figure 3; Fisher exact $p = 1.6 \times 10^{-7}$, odds ratio 5.68).

The mechanism is structural rather than incidental. An abnormality flag is a threshold quantisation of a continuous measurement. Once a value lies outside its reference range, moving it further outside

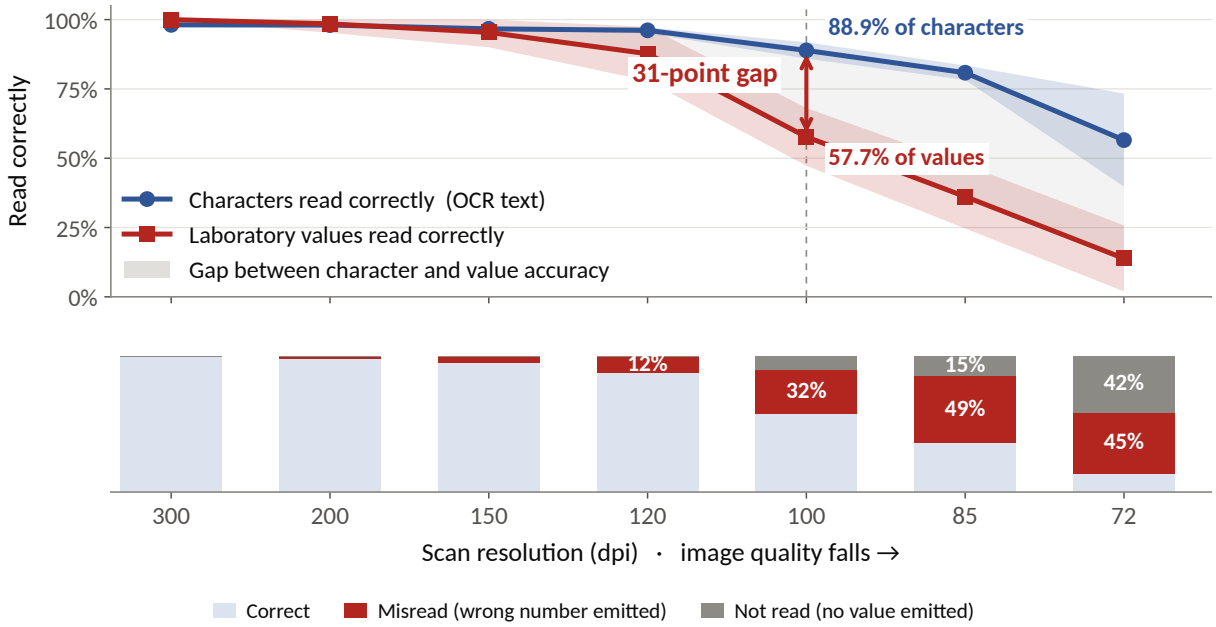


Figure 2: Character accuracy and value accuracy across the resolution sweep. Each point is the mean over 10 independent documents and bands are ± 1 standard deviation between documents, not confidence intervals. The lower panel decomposes each resolution into values read correctly, values misread, and values not read. At 100 dpi the two measures differ by 31 percentage points.

Table 3: What corruption did to the abnormality flag, for all 188 misread values in Experiment 2. Categories are mutually exclusive. The silent share computed within each document rather than pooled gives a mean of 48.2 percent (SD 12.7, range 23 to 64 across the 10 documents).

Outcome	Count	Share of corruptions (%)
Flag unchanged (silent)	89	47.3
Normal value flagged abnormal	53	28.2
Flag changed, wrong direction or severity	41	21.8
Abnormal value read as normal	5	2.7
Total	188	100.0

does not change which side of the threshold it falls on. Corruption that crosses a threshold is detected by the flagging layer; corruption that drives an already abnormal value deeper into abnormality is invisible by construction.

Decimal-shift errors make the same point quantitatively. Of 188 corruptions, 102 (54.3%) were decimal shifts, all of them upward. Tenfold shifts ($n = 87$) were silent in 42.5% of cases, while hundredfold shifts ($n = 15$) were silent in 86.7%. The larger error is the more invisible one.

Troponin I illustrates the consequence. Fourteen of its fifteen corrupted readings were hundredfold overstatements, and every one retained the *Critical* flag: 0.07 ng/mL read as 7.0, 0.82 read as 82.0. A borderline result and a result two orders of magnitude higher are indistinguishable downstream.

4.5 A plausibility bound helps, and fails systematically

The plausibility bound rejected 104 of the 188 corruptions (55.3%) and rejected none of the 636 correct values. The absence of false positives is guaranteed by the generator, which samples truth values strictly inside the envelope, and is not an estimate of a real false-positive rate.

Of the 89 silent corruptions, the bound recovered 33 (37.1%), leaving 56 that neither the flagging layer nor the plausibility check would surface.

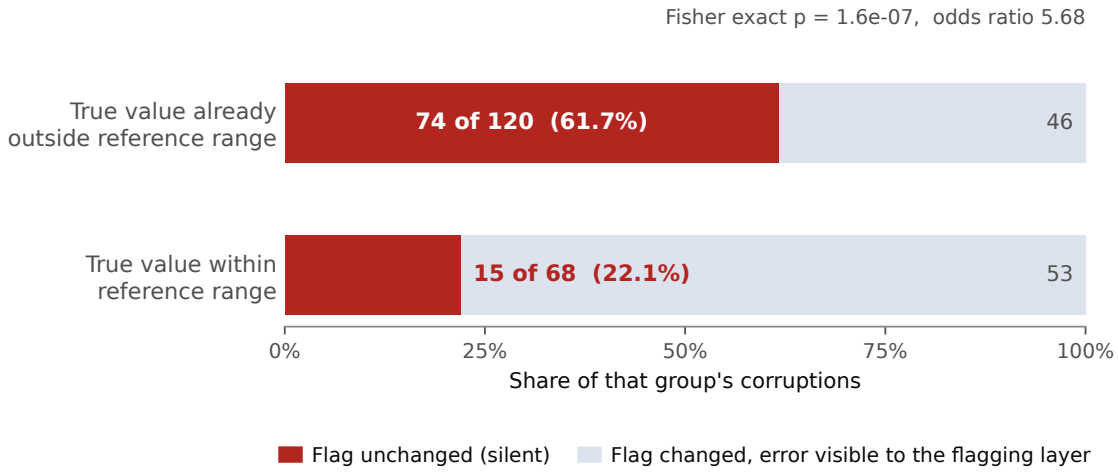


Figure 3: Corruption of a value that was already outside its reference range is far more likely to leave the abnormality flag unchanged. Bars show, within each group, the share of that group's corruptions that were silent. All 188 corruptions from Experiment 2 are included and the two groups are disjoint.

Table 4: Plausibility-bound detection rate per analyte, over the corruptions observed for that analyte in Experiment 2. Analytes whose plausible envelope is wide relative to their reference range, or whose envelope is open at the relevant end, are not protected.

Analyte	Corruptions	Detected (%)
Potassium	32	96.9
Haemoglobin	33	90.9
Sodium	6	83.3
Platelets	16	75.0
HbA1c	4	75.0
Creatinine	27	59.3
Blood urea nitrogen	7	42.9
White cell count	18	16.7
LDL cholesterol	8	12.5
eGFR	8	0.0
Fasting glucose	8	0.0
Ferritin	6	0.0
Troponin I	15	0.0

Per-analyte detection ranges from 96.9% for potassium to zero for troponin I, ferritin, fasting glucose and eGFR (Table 4). We tested whether detection is predicted by the ratio of plausible envelope width to reference range width. The association is in the expected direction and does not reach significance at this sample size (Spearman $\rho = -0.48$, $p = 0.09$, $n = 13$), and eGFR contradicts it: its envelope is the narrowest of the panel relative to its reference range, yet no corruption was detected, because the envelope's lower bound is zero and every observed eGFR corruption moved the value downward. Detectability depends on where a corrupted value lands relative to an envelope boundary, which is not captured by width alone.

4.6 A language model extractor repaired nothing

We replaced the deterministic parser with a language model extractor (Llama 3.2 3B via Ollama, temperature 0) and ran both over identical OCR text across 12 conditions at two repeats. In four of the 24 runs the model returned no values at all after emitting malformed JSON (`rotate_5.0` twice, `noise_30` twice). Over the 260 readings from the 20 runs that returned values, both extractors were correct on the

same 188: zero repairs and zero new errors.

The corruption is established before either extractor sees the text, and a model reading the same damaged characters reproduces the same damaged number.

5 Discussion

The practical consequence is that character-level accuracy does not bound value-level risk in a pipeline of this shape. A system reporting 88.9% character accuracy was delivering wrong values for roughly a third of the panel, and for nearly half of those the downstream alerting layer could not tell.

Threshold quantisation is common in clinical software, because reference ranges and severity bands are how laboratory results are presented. The finding here is that quantisation is also an information-destroying step placed immediately after an error-prone one. Any check that operates on the flag rather than the value inherits a blind spot whose size depends on how abnormal the true value was. The patients whose values are furthest outside the reference range are the patients whose corrupted readings are hardest to detect.

The plausibility bound is worth implementing and is not sufficient on its own. It is cheapest where it is least needed, for analytes like potassium whose physiological envelope is narrow, and it provides nothing for analytes like troponin whose plausible range spans several orders of magnitude.

Two mitigations follow from the mechanism rather than from the specific numbers. Carrying a magnitude check alongside the flag, for example a comparison against the range printed on the document itself rather than a hardcoded range, would detect errors the flag cannot. Treating unread and misread differently matters as well: an unread analyte is an absence a caller can notice, while a misread one is a confident wrong answer.

6 Limitations

The documents are synthetic. They carry exact ground truth, which is what makes the measurement possible, and they are not real clinical documents. Nothing here establishes a rate for any deployed system, and no clinician reviewed any output.

One OCR engine was used, at one page segmentation mode, with one document layout and one panel of 13 analytes. Every quantity we report is conditional on that stack.

Resolution and pixels per glyph are entangled, since body text varies between 8.5 and 10 point across the generated documents. A reader could reasonably hold that character size in pixels, rather than nominal resolution, is the operative variable, and this design cannot separate them.

The plausibility envelope is our construction. Its zero false-positive rate is an artefact of a generator that samples inside it.

The degradation sweep in Experiment 1 and the extractor comparison use a single document. Only the resolution sweep is replicated across 10 documents, and only the resolution results carry between-document dispersion.

We did not measure whether a corrupted value changes a clinical recommendation. The propagation of a corrupted value into generated text was observed on one document and is not quantified here.

7 Conclusion

Character accuracy and value accuracy come apart under degradation, and the gap is widest exactly where the clinical stakes are highest. Nearly half of the value corruptions we observed preserved the abnormality flag, and corruption of an already-abnormal value was more than five times as likely to be silent as corruption of a normal one, because a threshold cannot represent how far past it a value has moved. A plausibility bound recovers some of this and fails systematically for the analytes with the widest physiological ranges. Substituting a language model for a deterministic parser changed nothing,

because the information was lost before either could read it. Knowing where a number came from is not the same as knowing it was read correctly.

Data and code availability

The experiment scripts, generated documents, truth files, raw OCR output and all result tables are available in the project repository. Synthetic data are used throughout; no real patient information was processed at any stage.

Institutional review board

The research does not require IRB approval. It uses only synthetic documents generated for this study and involves no human subjects, human tissue or identifiable personal data.

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